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A checklist of butterfly diversity in Bhagwan Birsa Biological Park, Ormanjhi, Ranchi, Jharkhand

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Abstract- Butterflies, with their diverse colours and patterns, serve as valuable indicators of ecosystem health. This study presents a comprehensive checklist of butterfly species observed in Bhagwan Birsa Biological Park, Ormanjhi, Ranchi, Jharkhand, spanning the period from January, 2021 to January, 2023. The study area, situated at GPS coordinates 23°27'57"N, 85°27'20"E, was surveyed using the direct encounter method. A total of 86 butterfly species from six families were documented. This checklist provides essential information for understanding the local butterfly diversity, contributing to ongoing conservation efforts.

Key words: Butterfly Diversity, Bhagwan Birsa Biological Park, Checklist, Direct Encounter Method, Conservation

INTRODUCTION

Butterflies, with their intricate ecological roles, are vital components of biodiversity.[1] This study focuses on documenting the butterfly diversity in Bhagwan Birsa Biological Park, Ormanjhi, Ranchi, Jharkhand, providing valuable insights into the local ecosystem. Butterflies are not only aesthetically appealing but also function as key pollinators, contributing to the overall health of ecosystems.[2] Additionally, the classification and legal protection of butterfly species in India, as outlined by Mohanraj & Veenakumari (1996)[3], underscore the importance of conserving these species.

Study Area:

The study area is located at GPS coordinates 23°27'57" N, 85°27'20" E, within Bhagwan Birsa Biological Park, Ormanjhi, Ranchi, Jharkhand. The park encompasses diverse habitats, offering a suitable environment for a variety of butterfly species. The area of the park is 104 hectares.

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METHODOLOGY

The study employed a random encounter method,[4] encompassing random surveys conducted during various times of the day, including morning and night. The survey involved photographing butterflies, and subsequent identification was achieved by matching the captured images with reference material from the website www.ifoundbutterflies.org[5] and the book by Isaac Kehimkar (2008)[6]. This approach facilitated a comprehensive and systematic documentation of butterfly species encountered during the study period. The survey period extended from January, 2021 to January, 2023.

RESULTS

A total of 86 butterfly species belonging to six families were recorded during the study. The families include

Hesperiidae, Lycaenidae, Nymphalidae, Papilionidae, Pieridae, and Riodinidae. The detailed information on the identified species is presented in Table 1 through the checklist.





Table 1- A checklist of Butterfly Species

No. of Species	Common Name	*Scientific Name*	WPAI Status
Hesperiidae			
1	Common Branded Redeye	*Matapa aria*	
2	Grass Demon	*Udaspes folus*	
3	Rice Swift	*Borbo cinnara*	
4	Small Branded Swift	*Pelopidas mathias*	
5	Tricolour Pied Flat	*Coladenia indrani*	
6	Water Snow Flat	*Tagiades litigiosa*	
7	Asian Grizzled Skipper	*Spialia galba*	
8	Brown Awl	*Badamia exclamationis*	
9	Common Banded Awl	*Hasora chromus*	
10	Plain-Palm-Dart	*Cephrenes acalle*	
Lycaenidae			
11	Bright Sunbeam	*Curetis bulis*	
12	Common Cerulean	*Jamides celeno*	
13	Common Flash	*Rapala nissa*	
14	Common Red Flash	*Rapala iarbus*	
15	Slate Flash	*Rapala manea*	
16	Common Pierrot	*Castalius rosimon*	Sch. I
17	Dark Grass Blue	*Zizeeria karsandra*	
18	Dark Oak Blue	*Arhopala rama*	
19	Forget-Me-Not	*Catochrysops strabo*	
20	Indigo Flash	*Rapala varuna*	Sch. II
21	Lesser Grass Blue	*Zizina otis*	
22	Pale Grass Blue	*Pseudozizeeria maha*	

23	Plains Cupid	*Chilades pandava*	
24	Red Pierrot	*Talicada nyseus*	
25	Yamfly	*Loxura atymnus*	
26	Lineblue	*Nacaduba sinhala*	
27	Zebra Blue	*Leptotes plinius*	
28	Peacock Royal	*Tajuria cippus*	
29	Mandarin Blue	*Charana mandarinus*	
30	Little Tiger Peirrot	*Tarucus balkanica*	
31	Metallic Cerulean	*Jamides alecto*	Sch. II
32	Purple Leaf Blue	*Amblypodia anita*	Sch. II
34	Gram Blue	*Euchrysops cnejus*	Sch. II
Nymphalidae			
35	Baronet	*Symphaedra nais*	
36	Blue Pansy	*Junonia orithya*	
37	Blue Tiger	*Tirumala limniace*	
38	Chocolate Pansy	*Junonia iphita*	
39	Colour Sergeant	*Athyma inara*	
40	Commander	*Moduza procris*	
41	Common Baron	*Euthalia aconthea*	Sch. II
42	Common Castor	*Ariadne merione*	
43	Common Crow	*Euploea core*	
44	Common Evening Brown	*Melanitis leda*	
45	Common Four-Ring	*Ypthima huebneri*	
46	Common Lascar	*Pantoporia hordonia*	
47	Common Leopard	*Phalanta phalantha*	
48	Common Palmfly	*Elymnias hypermnestra*	
49	Common Sailer	*Neptis hylas*	
50	Common Sergeant	*Athyma perius*	
51	Gaudy Baron	*Euthalia lubentina*	Sch. IV
52	Danaid Eggfly	*Hypolimnas misippus*	Sch. I
53	Great Eggfly	*Hypolimnas bolina*	
54	Great Four-Ring	*Ypthima cantliei*	
55	Grey Count	*Tanaecia lepidea*	Sch. II
56	Grey Pansy	*Junonia atlites*	
57	Lemon Pansy	*Junonia lemonias*	
58	Lilacine Bushbrown	*Mycalesis francisca*	
59	Orange Staff Sergeant	*Athyma cama*	
60	Peacock Pansy	*Junonia almana*	
61	Plain Tiger	*Danaus chrysippus*	
62	Short-Banded Sailer	*Phaedyma columella*	
63	Striped Tiger	*Danaus genutia*	Sch. I
64	Tawny Coster	*Acraea terpsicore*	
65	Yellow Pansy	*Junonia hierta*	
66	Painted Lady	*Vanessa cardui*	
67	Malayan Nawab	*Charaxes moori*	
68	Tawny Rajah	*Charaxes bernardus*	Sch. II
Papilionidae			
69	Blue Mormon	*Papilio polymnestor*	
70	Common Banded Peacock	*Papilio crino*	

71	Common Jay	*Graphium doson*	
72	Common Mime	*Papilio clytia*	Sch. I
73	Common Mormon	*Papilio polytes*	
74	Common Rose	*Pachliopta aristolochiae*	
75	Lime Swallowtail	*Papilio demoleus*	
76	Tailed Jay	*Graphium agamemnon*	
77	Spot Swordtail	*Graphium nomius*	
Pieridae			
78	Common Grass Yellow	*Eurema hecabe*	
79	Common Gull	*Cepora nerissa*	
80	Indian Jezebel	*Delias eucharis*	
81	Indian Wanderer	*Pareronia hippia*	
82	Lemon Emigrant	*Catopsilia pomona*	
83	Mottled Emigrant	*Catopsilia pyranthe*	
84	Psyche	*Leptosia nina*	
85	Pioneer	*Belenois aurota*	
Riodinidae			
86	Plum Judy	*Abisara echerius*	

DISCUSSION

The observed diversity of butterfly species in Bhagwan Birsa Biological Park highlights the importance of this area for butterfly conservation. The checklist serves as a foundational reference for future studies and conservation initiatives, emphasizing the need to protect and preserve the diverse habitats within the park.

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CONCLUSION

This study contributes a comprehensive checklist of butterfly species in Bhagwan Birsa Biological Park, Ranchi, Jharkhand, shedding light on the rich biodiversity of the area. The findings underscore the significance of the park as a habitat for diverse butterfly species, emphasizing the importance of conservation measures.

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